

PI Session: Recurrent Neural Networks

This 90-minute PI session aims to provide a comprehensive understanding and clarification of machine learning concepts. The session is divided into three main parts:

- Doubt resolution for Neural Networks (NN) and Convolutional Neural Network (CNN) modules.
- Discussion on industry applications of Recurrent Neural Networks (RNN).
- Discussion on interview questions regarding NN, CNN and RNN.

Before coming to the objectives of this session, let's update you with learners' learning status. The learners have completed the following topics:

- Python for Data Science
- Data Visualization using Python
- Exploratory Data Analysis (EDA)
- Basic and Advanced SQL
- Inferential Statistics
- Hypothesis Testing
- Linear Regression
- Logistic Regression
- Naive Bayes
- Model Selection
- Advanced Regression
- Decision Tree
- Random Forest
- Boosting
- Unsupervised Clustering
- PCA
- Neural Network (NN)
- Convolutional Neural Network (CNN)

Note:

This week learners will begin learning about Recurrent Neural Networks (RNN). Live sessions are conducted to cover conceptual topics and demonstrations. The agenda for these sessions on RNN is provided below. When discussing industry applications, please avoid topics from these agendas that learners may not yet understand.

Recurrent Neural Networks Theory Session (Basics)

Since learners are just starting with Recurrent Neural Networks this week, emphasize the industry applications of RNN and practical insights based on your experiences.

Objectives:

- Clarify doubts regarding previous modules on NN and CNN.
- Understand industry applications of RNN.
- Discuss frequently asked interview questions and resolve doubts on ML concepts (NN, CNN and RNN).

Instructions for the Instructor:

- Review the key concepts, common doubts, and frequently asked interview questions for each concept. Be ready with practical examples and clarifications.
- **Encourage learners to actively participate by asking questions and sharing their experiences.**
- Ensure each segment stays within the allocated time to cover all planned topics.
- Address specific queries and provide additional insights.
- Prepare additional materials or references for learners who might need further study on specific topics.

Expectations from the Instructor:

- As this session focuses on clearing learners' doubts, ask learners about their doubts on specific topics. Address the most frequently asked questions with clear examples.
- Avoid covering points related to RNN that will be discussed in the upcoming conceptual or demonstration sessions.
- Instructors can use their own knowledge of common interview questions and share their experiences to guide the learners. For reference, we have provided a few questions from each topic. Feel free to expand on these with examples and scenarios from your professional experience.
- Address at least 4-5 common doubts or interview questions from the detailed breakdown during the session. Ensure no repetition of topics already covered by student questions.

Here is the breakdown of the agenda:

1. Introduction and Session Overview (5 mins)
2. Doubt Resolution on previous modules: Neural Networks ,CNN (25 mins)
3. Industry Applications: Recurrent Neural Networks (RNN) (30 mins)
4. Interview Questions Discussion on previous ML Concepts (30 mins)

Detailed Breakdown:

1. Introduction and Session Overview (5 mins)

- Brief introduction and purpose of the session.
- Outline the structure of the session and the key concepts to be covered.

2. Doubt Resolution on previous modules: Neural Networks ,CNN (25 mins)

- **Neural Network**
 - Quick recap of critical topics: layers, neurons, activation functions.
 - Address common issues such as understanding architecture and designing NNs.
 - Address common issues such as flow of data and activation functions.
 - Address common issues such as functionality and limitations.
 - Address common issues such as gradient descent and error calculation.
 - Address common issues such as optimization and model tuning.
 - Quick recap of modifications for different applications.
 - Address common issues such as customization and performance improvement.
- **Q&A and Discussion**
 - Open the floor for questions from learners to clarify doubts.

3. Industry Applications: Recurrent Neural Networks (RNN) (30 mins)

- **Industry Applications**
 - Briefly explain the purpose of RNN in prediction and sequence tasks.
 - Identify industries where RNN models are frequently used (e.g., finance, healthcare, retail).
 - Discuss common variables typically used in these industries for RNN analysis.
 - Explain how RNN models help in making predictions and influencing decisions within these industries.
 - Discuss when to use feedforward networks, convolutional networks, or recurrent networks for industry problems.
- **Q&A and Discussion**
 - Open the floor for questions from learners to clarify doubts and discuss specific industry scenarios.

4. Interview Questions Discussion on previous ML Concepts (30 mins)

- **Neural Networks**
 - How do you choose the right architecture for a neural network?
 - What are some common pitfalls in training neural networks?
- **Recurrent Neural Networks**
 - How do you handle vanishing and exploding gradients in RNNs?
 - What is the purpose of LSTM and GRU in RNNs?
- **Open Q&A**

- Facilitate an open Q&A session for any remaining doubts from the overall session.
- Provide additional resources and tips for further study if needed.